

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 1: FIND-S Algorithm

Aim

To implement and demonstrate the **FIND-S algorithm** for finding the most specific hypothesis from the given training data.

Procedure

1. Define the training dataset.
2. Initialize the hypothesis with the most specific values (0).
3. Consider only the positive training examples.
4. Update the hypothesis by replacing mismatched attributes with ?.
5. Display the final hypothesis.

Program

```
data = [  
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],  
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],  
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],  
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']  
]  
  
h = ['0'] * 6  
  
for row in data:  
    if row[-1] == 'Yes':  
        for i in range(6):  
            if h[i] == '0':  
                h[i] = row[i]  
            elif h[i] != row[i]:  
                h[i] = '?'  
  
print("Final Hypothesis:", h)
```

Output

Final Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

Result

Thus, the FIND-S algorithm was implemented successfully, and the most specific hypothesis was obtained.